



Swirls of light in the CMB

These swirls of light at the edge of the universe may point to an aeon before ours
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Hubble dating Methuselah

How the HST was used to provide a birth certificate for one of the oldest stars known
<https://digit.in/oct19-04>

old, give or take 20 million years. This would mean that somehow, the Methuselah star was older than the universe. Observations from the Hubble space telescope help determine the age of the star better, giving it a kind of a birth certificate. The precise measurements indicated a minimum age of 13.66 billion years, which makes it more compatible with the age of the universe. However, that still means that the star was formed very soon after the universe came into being.

BLACK HOLES

According to conventional theories, a black hole is formed when a very massive star close to the end of its life cycle, collapses in on itself. However, there are a number of super massive black holes observed, that are in such high redshifts, that they must have formed within 800 million years after the Big Bang. In a way, although they are incredibly ancient, these black holes can also be called “young” considering how early they were formed. In 2019, a team of astronomers announced the discovery of 83 black holes, that were formed within 800 million years of the Big Bang. These black holes were identified through glowing gases in the accretion disks of infalling material. The black holes were found after a survey of five years. The finding is a challenge to conventional theories of black hole formation.

Astrophysicists from Western University might have a solution. According to their explanation, a large number of supermassive black holes were formed very early in the universe, and their formation suddenly stopped after their numbers hit a particular limit, because of the amount of material and energy available. They have produced a model that could explain the formation of supermassive black holes, so early in the universe. The explanation proposes a novel way for supermassive black holes to form, through a direct collapse, as against the collapsing remains of a dying star. Shantanu Basu, one of the authors of the study explains, “This is indirect observation-

al evidence that black holes originate from direct-collapses and not from stellar remnants. Supermassive black holes only had a short time period where they were able to grow fast and then at some point, because of all the radiation in the universe created by other black holes and stars, their production came to a halt. That’s the direct-collapse scenario.”

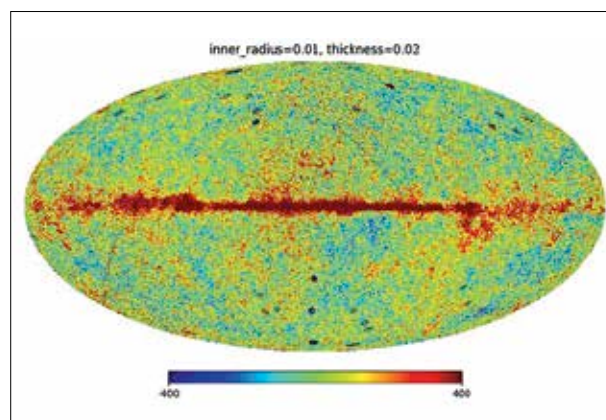
CYCLIC COSMOLOGY

One of the most dramatic findings of Stephen Hawking’s was that black holes are not entirely black. Even black holes can radiate, because of quantum effects near the event horizon. Low temperature photons and neutrinos are radiated from black holes. As the universe expands and cools, over a

freeze or heat death, as it approaches a state of maximum entropy. It may also collapse into itself in a big crunch. According to the Big Rip theory, the universe will fall apart into elementary particles and radiation. According to a theory proposed by Sir Roger Penrose, the universe is cyclic. That means that the continual expansion farms a singularity, which explodes again.

The theory is called Conformal cyclic cosmology (CCC), and is within the framework of general relativity. Penrose expounded on the concept in his 2010 book, *Cycles of Time: An Extraordinary New View of the Universe*. According to CCC, the universe is like a phoenix, emerging from its own remains time and again, in periods that Penrose refers to as aeons.

If that is the case, then the remnants of the Hawking Radiation from the black holes should be visible in the cosmic microwave background. Penrose claims to have found such evidence in the CMB of our current universe, locations known as Hawking Points or Hawking Holes. Using data



Hawking Points in the CMB are marked with the black circles

period of 10^{100} years, the supermassive black holes at the centers of galaxies should start glowing. This anticipated glow is very faint, less than about one ten millionth of a degree above absolute zero. Yet, if cosmological time scales are compressed (think looking at a fast forwarded universe), then even this faint glow would appear as an explosion.

Now we have to take a short detour on the currently accepted model of how the universe formed and evolved, known as cosmic inflation. Essentially, there is a period of inflation or expansion after the big bang. There are several theories on the ultimate fate of the universe. The universe may just expand forever resulting in a big

from the BICEP 2 observatory, Penrose claims to have identified 30 Hawking Points, which may be the telltale signs of black holes in a previous universe. According to Penrose, these are the marks of the final evaporation of supermassive black holes, in an aeon previous to the one that we live in. Complex data processing techniques were required to isolate these points, and other scientists claim that the whole effort may just be an overinterpretation of random noise. If the supposed anomalies are in fact there, then explaining them away or reconciling them with the conventional models of the universe would be extremely difficult. The finding, and the theory are both controversial.